

MOHAMED JRIDI

Düsseldorf, Germany

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With experience in Software IoT and Backend, I strive to deliver high-quality software solutions that drive business value. I enjoy working in teams with a problem solving mindset, both solving software and development cycle bottlenecks. I am looking for challenging opportunities with companies that look for excellence.



Professional Experience

Freelancer - Düsseldorf

09/2024 - Ongoing

- Built websites for small businesses.
- Built projects using AI Agents, including ChatBots and Browser Extensions (Plugins).
- Mastered the full AI Agents development lifecycle, including function calling with **MCP**, Tool chaining, guardrails and **Context Engineering** with **memory**.
- Achieved significant progress in the German language.

Lead Software Engineer - Livello Gmbh, Düsseldorf

11/2023 - 08/2024

Led a team of 5 developers in developing and modernizing a full-stack (**Python, Java, JS/TS**), cloud-to-edge product. Drove key initiatives that directly expanded the company's market presence and solidified its technical foundation.

- Leadership: Managed the **agile process** from concept to deployment, fostering team growth through **mentoring, code reviews, pair-programming**, and hands-on technical guidance. Led the adoption of Behavior-Driven Development (**BDD**), collaborating with Biz and QA. This process **eliminated ambiguity in requirements**, accelerated development cycles, and enhanced end-product quality by ensuring **features met defined acceptance criteria** from the outset.
- Legacy System Modernization: Worked on a project to **refurbish** the entire **legacy** device fleet with our latest IoT capabilities, **protecting a vital revenue stream** and significantly extending asset lifetime.
- Product Development:
 - Achieved **certification** for a new international payment gateway, a key enabler for securing new clients and successfully expanding into **new European markets**.
 - Engineered an **RFID** Electronic Shelf Label system that automates in-store price visualization, eliminating manual errors and operational overhead.
 - In-house OTA for releases and software updates of the remote devices using Ansible.
- Engineering Foundation: Slashed **technical debt by 90%** through taking initiative, boosting long-term maintainability. Automated infrastructure and **CI/CD pipelines** to accelerate delivery and improve reliability.
- Client Support: Around the clock support for client through the phone and worked on developing a **new ticketing system** to speed up the client support process and solve issues faster.

Engineered and modernized a cloud-edge platform, focusing on a scalable architecture with a robust deployment processes, and high-reliability systems for secure IoT data. Working extensively with **Python**, **Java** and **Javascript**.

- Refactoring and Modernization:
 - Designing an **event-driven** architecture using **async** Python and **Vert.x** to enhance performance and **scalability**.
 - Orchestrated the full **migration** from **Google IoT Core** to a **ClearBlade IoT Core** with zero downtime.
- Automated Fleet Deployment & Management: Built a robust, automated device provisioning system using **Ansible**, to manage the **configuration**, **deployment**, and **updates** for a large-scale device fleet remotely.
- Extending Supported Devices:
 - Implemented low-level integration for **ZVT** and **MDB** payment protocols, ensuring compliant reliable financial transactions.
 - Engineered network resilience through backup Wi-Fi Dongle and Quectel LTE module for constant uptime.
- Enhanced System Observability for Support: Designed and deployed a comprehensive **monitoring** stack using **Prometheus** and **Grafana** on the edge devices, providing **real-time health metrics** and slashing mean time to resolution (**MTTR**) for bugs.

Internship - Livello GmbH, Düsseldorf

09/2021 - 03/2022

Software Developer in IoT, taking care of maintaining software, resolving bugs and a focus on solving a payment process speed issue thus reducing it by 70% to 80%.

Bachelor Thesis - WorldSoft Group, Tunis

02/2018 - 06/2018

Full-stack Developer, I developed a Java/Angular **web application** for B2B.

Education

- **Master's Degree - Engineering studies, branch of Internet of Things Systems and Services (IoSyS), ESPRIT School of Engineering**
- **Exchange Student at Fachhochschule Schmalkalden, Thuringen**
 - Program: 6 Months
 - Software Engineering/Development program
 - Scored from 1 to 1,7
- **Bachelor's Degree in Fundamental Computer Science**

Technical Skills

Programming & Languages

Python, Java, C/C++ (Arduino IDE), Javascript/TypeScript

Frameworks/Libraries: Spring Boot, Vert.x, Node.js, FastAPI, FastMCP, Google genai, aiogram

Cloud & IoT Platforms

Google Cloud Platform (GCP), Docker, Kubernetes, OAuth2, ClearBlade

DevOps & Tooling

CI/CD, Ansible, Prometheus, Grafana, Github, GitLab

Message Brokers: RabbitMQ, Mosquitto

Databases

MongoDB, Redis, PostgreSQL

Systems & Networking

Linux System Administration, ARM Architecture, OpenVPN, Arch Linux

Protocols: MQTT, AMQP, BLE, ZVT, MDB, LTE, NFC, RFID

Development Boards: Raspberry Pi 4B+/Pico, Arduino, STM32

Methods & Architecture

Agile/Scrum, Microservices, Event-Driven Architecture, Reactive Programming, BDD, TDD, UML

Additional Skills

Artificial Intelligence

LLM APIs (Gemini, DeepSeek, Mistral), MCP, PydanticAI, Context Engineering, RAG

Tools & Development

Jira, Confluence, VS Code, Next-Gen AI Code Tools

Languages

- **Arabic:** Mother tongue
 - **German:** Beginner B1
 - **English:** Fluent
 - **French:** Fluent
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Hobbies

- Technical/Non-Technical Reading
- Cooking
- Calisthenics